

Antonio Lang

Reno, NV

github.com/antoniolang1107 | [linkedin.com/in/antonio-lang/](https://www.linkedin.com/in/antonio-lang/)

Aspiring engineer passionate to solve problems concerning data privacy and the right to ownership. National Science Foundation S-STEM CREATE scholar, served as president of UNR's Circle K club, and eager to apply my skills and knowledge to solve global challenges.

Work Experience

Desert Research Institute

Reno, Nevada

Full-Stack Developer

April 2024 - July 2026

- Developed web platform (C#, ASP.NET) to simplify environmental comparisons between regions (ArcGIS, ChartJS, Potree)
- Assisted on variety of projects including GUI development, API integration, and machine learning model training
- Prototyped RAG system for internal scientific reports and n8n workflow automation
- Developed lab task and data management software with React frontend, Python Django backend; managing project and UI design
- Trained dust visibility machine learning model for environmental research, greatly exceeding project requirements

Tesla

Sparks, Nevada

Production Engineering Intern

Sept 2023 - Dec 2023

- Data management in CMMS (correcting 1300 discrepancies) supporting maintenance and reliability teams for root cause failure analysis
- Developed Python pipelines to automate spare part strategy process to reduce human error and guide decision making for multi-million dollar warehouse inventory, increasing task efficiency by 70%
- Followed test-driven development for pipeline software and maintained documentation for various technical backgrounds

UNR High Performance & Data Intensive Computing Lab

Reno, Nevada

Machine Learning Researcher

Jun 2023 - Aug 2023

- Developing novel data-decoupled federated learning (FL) framework
- Performed standardized testing on a compute cluster (leveraging Cassandra) using numerous FL frameworks

UNR High Performance & Data Intensive Computing Lab

Reno, Nevada

Blockchain Researcher

Jun 2022 - Aug 2022

- Conducted research on creating a decentralized, nonblocking protocol for cross-blockchain transactions
- Developed Python-based PoW blockchains and implemented the commit protocol on top of the blockchains to further the research project

Education

University of Nevada, Reno (UNR)

Reno, Nevada

M.S. of Computer Science & Engineering (Machine Learning focus); GPA: 4.0

Jan 2024 - May 2025

- Adversarial Machine Learning, Big Data and Distributed Systems, Bayesian Statistics, Bayesian and Frequentist Hypothesis Testing
- US Automobile Accident Exploratory Data Science

University of Nevada, Reno

Reno, Nevada

B.S. of Computer Science & Engineering (Big Data, Mathematics minors); GPA: 3.9

Aug 2020 - Dec 2023

- NSF "Creating Retention and Education for Academically Talented Engineers" Scholar; program representative at AAAS National S-STEM Conference

Projects

Self-Hosted Home Server

Reno, NV

Personal Project

Present

- Proxmox server to self-host services for data privacy and ownership
- PiHole VPN, Nginx Proxy Manager, Ollama for self-hosting LLM-based services

Model-Decoupled Federated Learning: Exploring Database Adaptability and Neural Architectural Dynamics in Federated Learning

Reno, NV

University of Nevada, Reno

August 2024

- DOI: 10.13140/RG.2.2.26105.68963

Technical Skills

Programming

Python (Pandas, NumPy, Scikit-learn, matplotlib), R, SQL, Postgres, Docker, Hadoop, MapReduce, Spark, React, Vue.js

Office and Project

Shell (Bash/Zsh), LaTeX (Overleaf/R Markdown), Microsoft Excel, git/GitHub, REST API

Soft Skills

Leadership, Teamwork, Time Management, Problem-solving, Presentation, Communication